

MATH 152 — Part 4

Rolle's Theorem, Mean Value Theorems, and Integration

Detailed Notes with Tips and Examples

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4.1 Rolle's Theorem

Statement of the Theorem

If a function $f(x)$ satisfies the following three conditions:

1. **Continuous** on the closed interval $[a, b]$
2. **Differentiable** on the open interval (a, b)
3. **$f(a) = f(b)$** (equal values at endpoints)

Then there exists **at least one** point $x = \xi \in (a, b)$ such that:

$$f'(\xi) = 0$$

Geometrical Interpretation

Rolle's Theorem states that if a smooth curve has the same y-value at two points, then somewhere between those points, the tangent to the curve must be **horizontal** (parallel to the x-axis).

Key Insight: There is at least one point where the tangent is parallel to the x-axis.

Proof of Rolle's Theorem

Case 1: $f(x) = 0$ for all $x \in [a, b]$

- Then $f'(x) = 0$ for all x in (a, b)
- The theorem is trivially true.

Case 2: $f(x) \neq 0$ for some $x \in [a, b]$

Since f is continuous on $[a, b]$, by the Extreme Value Theorem:

- f attains a **maximum** M at some point $\xi_1 \in [a, b]$
- f attains a **minimum** m at some point $\xi_2 \in [a, b]$

Since $f(x) \neq 0$, at least one of M or m is not zero.

Suppose $M \neq 0$ and $f(\xi) = M$ for some $\xi \in (a, b)$.

For any h (small enough), $f(\xi + h) \leq f(\xi)$

For $h > 0$:

$$[f(\xi + h) - f(\xi)]/h \leq 0$$

Taking limit as $h \rightarrow 0^+$:

$$f'_+(\xi) \leq 0$$

For $h < 0$:

$$[f(\xi + h) - f(\xi)]/h \geq 0$$

Taking limit as $h \rightarrow 0^-$:

$$f'_-(\xi) \geq 0$$

Since f is differentiable at ξ , $f'_-(\xi) = f'_+(\xi) = f'(\xi)$

Therefore: $f'(\xi) \leq 0$ and $f'(\xi) \geq 0 \rightarrow f'(\xi) = 0$

If $m \neq 0$, a similar argument with the minimum gives $f'(\xi) = 0$.

This completes the proof.

Example 1: Verifying Rolle's Theorem

Problem: Verify Rolle's Theorem for $f(x) = x(2x - 1)$ on $[0, 1/2]$

Solution:

Step 1: Check conditions

$$f(x) = 2x^2 - x$$

1. **Continuity:** Polynomial $\rightarrow$ continuous on $[0, 1/2]$ ✓
2. **Differentiability:** $f'(x) = 4x - 1 \rightarrow$ exists on $(0, 1/2)$ ✓
3. **Equal endpoints:**

$$f(0) = 0$$

$$f(1/2) = (1/2)(1 - 1) = (1/2)(0) = 0$$

$$f(0) = f(1/2) = 0 \quad \checkmark$$

All conditions satisfied.

Step 2: Find ξ where $f'(\xi) = 0$

$$f'(x) = 4x - 1$$

$$f'(\xi) = 0$$

$$4\xi - 1 = 0$$

$$\xi = 1/4$$

Step 3: Verify $\xi \in (0, 1/2)$

$$0 < 1/4 < 1/2 \quad \checkmark$$

Rolle's Theorem is verified!

Example 2: Rolle's Theorem Application

Problem: Show that the equation $x^3 - 3x + 1 = 0$ has at most one root in $(0, 1)$.

Solution:

Assume there are two roots a, b in $(0, 1)$ with $a < b$.

Then $f(a) = f(b) = 0$, where $f(x) = x^3 - 3x + 1$.

By Rolle's Theorem, $\exists \xi \in (a, b)$ such that $f'(\xi) = 0$.

$$\text{But } f'(x) = 3x^2 - 3 = 3(x^2 - 1)$$

$$f'(\xi) = 0 \rightarrow 3(\xi^2 - 1) = 0 \rightarrow \xi^2 = 1 \rightarrow \xi = \pm 1$$

But $\xi \in (0, 1)$, so ξ cannot be ± 1 . **Contradiction!**

Therefore, the equation has at most one root in $(0, 1)$.

Important Note

The condition $f(a) = f(b) = 0$ can be replaced with the more general condition:

$$f(a) = f(b) \quad (\text{they don't have to be zero})$$

Why? We can always shift the function vertically by subtracting $f(a)$.

4.2 The First Mean Value Theorem (Lagrange's MVT)

Statement of the Theorem

If a function $f(x)$ satisfies:

1. **Continuous** on the closed interval $[a, b]$
2. **Differentiable** on the open interval (a, b)

Then there exists **at least one** point $x = \xi \in (a, b)$ such that:

$$f'(\xi) = [f(b) - f(a)] / (b - a)$$

Geometrical Interpretation

The Mean Value Theorem states that there is at least one point where the tangent is **parallel to the chord** joining $(a, f(a))$ and $(b, f(b))$.

Key Insight: The average rate of change equals the instantaneous rate of change at some point.

Proof of the First MVT

Construct the function:

$$G(x) = f(x) - y$$

where y is the equation of the chord AB joining $(a, f(a))$ and $(b, f(b))$.

Equation of chord AB:

$$y - f(a) = [f(b) - f(a)] / (b - a) \cdot (x - a)$$

$$y = [f(b) - f(a)] / (b - a) \cdot (x - a) + f(a)$$

Therefore:

$$G(x) = f(x) - [f(b) - f(a)] / (b - a) \cdot (x - a) - f(a)$$

Now, $G(x)$ is continuous and differentiable (since f is).

$$G(a) = f(a) - 0 - f(a) = 0$$

$$\begin{aligned} G(b) &= f(b) - [f(b) - f(a)] / (b - a) \cdot (b - a) - f(a) \\ &= f(b) - f(b) + f(a) - f(a) = 0 \end{aligned}$$

So $G(a) = G(b) = 0$.

By Rolle's Theorem, $\exists \xi \in (a, b)$ such that $G'(\xi) = 0$.

$$G'(x) = f'(x) - [f(b) - f(a)] / (b - a)$$

$$G'(\xi) = 0$$

$$f'(\xi) - [f(b) - f(a)] / (b - a) = 0$$

$$f'(\xi) = [f(b) - f(a)] / (b - a)$$

This completes the proof.

Relationship to Rolle's Theorem

If $f(b) = f(a)$, the MVT reduces to:

$$f'(\xi) = 0$$

This is exactly Rolle's Theorem!

So Rolle's Theorem is a **special case** of the First Mean Value Theorem.

Example 1: Verifying the MVT

Problem: Verify the MVT for $f(x) = x^2 - 3x + 1$ on $[0, 1]$

Solution:

Step 1: Check conditions

- f is a polynomial $\rightarrow$ continuous on $[0,1]$ ✓
- f is a polynomial $\rightarrow$ differentiable on $(0,1)$ ✓

Step 2: Find $f'(\xi)$

$$f'(x) = 2x - 3$$

$$f'(\xi) = 2\xi - 3$$

Step 3: Apply MVT formula

$$[f(1) - f(0)]/(1 - 0) = f'(\xi)$$

$$f(1) = 1 - 3 + 1 = -1$$

$$f(0) = 0 - 0 + 1 = 1$$

$$[f(1) - f(0)]/(1 - 0) = (-1 - 1)/1 = -2$$

$$\text{So: } 2\xi - 3 = -2$$

$$2\xi = 1$$

$$\xi = 1/2$$

Step 4: Verify $\xi \in (0, 1)$

$$0 < 1/2 < 1 \quad \checkmark$$

The MVT is verified!

Example 2: Application of MVT

Problem: Show that $|\sin b - \sin a| \leq |b - a|$ for all real a, b .

Solution:

If $a = b$, the inequality is trivially true ($0 \leq 0$).

Assume $a < b$. Let $f(x) = \sin x$.

f is continuous and differentiable everywhere.

By the MVT, $\exists \xi \in (a, b)$ such that:

$$\begin{aligned} [f(b) - f(a)]/(b - a) &= f'(\xi) \\ [\sin b - \sin a]/(b - a) &= \cos \xi \end{aligned}$$

Taking absolute values:

$$|\sin b - \sin a| = |\cos \xi| \cdot |b - a|$$

Since $|\cos \xi| \leq 1$:

$$|\sin b - \sin a| \leq |b - a|$$

This is a very useful inequality!

Example 3: Using MVT to Find c

Problem: Find the value(s) of c guaranteed by the MVT for $f(x) = x^3 - 2x$ on $[0, 2]$.

Solution:

$$\begin{aligned} f(0) &= 0 \\ f(2) &= 8 - 4 = 4 \end{aligned}$$

$$[f(2) - f(0)]/(2 - 0) = 4/2 = 2$$

$$\begin{aligned} f'(x) &= 3x^2 - 2 \\ f'(c) &= 3c^2 - 2 \end{aligned}$$

Set $f'(c) = 2$:

$$3c^2 - 2 = 2$$

$$3c^2 = 4$$

$$c^2 = 4/3$$

$$c = \pm 2/\sqrt{3}$$

Since c must be in $(0, 2)$:

$$c = 2/\sqrt{3} \approx 1.155 \in (0, 2)$$

Example 4: Existence of Root Using MVT

Problem: Show that the equation $2x^3 - 3x^2 + 1 = 0$ has at least one root in $(0, 1)$.

Solution:

$$\text{Let } f(x) = 2x^3 - 3x^2 + 1$$

$$f(0) = 0 - 0 + 1 = 1$$

$$f(1) = 2 - 3 + 1 = 0$$

Since $f(1) = 0$, $x = 1$ is a root.

But we need a root in $(0, 1)$.

Wait, $f(0) = 1$ and $f(1/2) = ?$

$$f(1/2) = 2(1/8) - 3(1/4) + 1 = 1/4 - 3/4 + 1 = 1/2$$

$$f(0) = 1, f(1/2) = 1/2, f(1) = 0$$

By the Intermediate Value Theorem, since $f(1/2) > 0$ and $f(1) = 0$, there must be a root between $1/2$ and 1 .

So the equation has a root in $(1/2, 1) \subset (0, 1)$.

4.3 The Second Mean Value Theorem (Cauchy's Generalized MVT)

Statement of the Theorem

If $f(x)$ and $g(x)$ are:

1. **Continuous** on the closed interval $[a, b]$
2. **Differentiable** on the open interval (a, b)
3. **$g(a) \neq g(b)$**

Then there exists **at least one** point $x = \xi \in (a, b)$ such that:

$$f'(\xi) / g'(\xi) = [f(b) - f(a)] / [g(b) - g(a)]$$

where $f'(x)$ and $g'(x)$ are not simultaneously zero.

Proof

Construct the function:

$$H(x) = f(x) - f(a) - \alpha[g(x) - g(a)]$$

where α is a constant.

$$\text{Choose } \alpha = [f(b) - f(a)]/[g(b) - g(a)]$$

Then:

$$H(a) = f(a) - f(a) - \alpha[g(a) - g(a)] = 0$$

$$H(b) = f(b) - f(a) - \alpha[g(b) - g(a)] = 0$$

By Rolle's Theorem, $\exists \xi \in (a, b)$ such that $H'(\xi) = 0$.

$$H'(x) = f'(x) - \alpha g'(x)$$

$$H'(\xi) = 0$$

$$f'(\xi) - \alpha g'(\xi) = 0$$

$$f'(\xi)/g'(\xi) = \alpha$$

Therefore:

$$f'(\xi)/g'(\xi) = [f(b) - f(a)]/[g(b) - g(a)]$$

This completes the proof.

Relationship to First MVT

If we set $g(x) = x$, then:

- $g'(x) = 1$
- $g(b) - g(a) = b - a$

The theorem becomes:

$$f'(\xi)/1 = [f(b) - f(a)]/(b - a)$$

$$f'(\xi) = [f(b) - f(a)]/(b - a)$$

This is exactly the First Mean Value Theorem!

So the First MVT is a **special case** of the Second MVT.

Example: Verifying Cauchy's MVT

Problem: Verify Cauchy's MVT for $f(x) = x^2$ and $g(x) = x^3$ on $[1, 2]$.

Solution:

Step 1: Check conditions

- Both are polynomials $\rightarrow$ continuous and differentiable $\checkmark$

Step 2: Compute values

$$f(1) = 1, f(2) = 4$$

$$g(1) = 1, g(2) = 8$$

Step 3: Apply formula

$$[f(2) - f(1)]/[g(2) - g(1)] = (4 - 1)/(8 - 1) = 3/7$$

Step 4: Find $f'(\xi)$ and $g'(\xi)$

$$f'(x) = 2x$$

$$g'(x) = 3x^2$$

$$f'(\xi)/g'(\xi) = 2\xi/(3\xi^2) = 2/(3\xi)$$

Step 5: Set equal and solve

$$2/(3\xi) = 3/7$$

$$14 = 9\xi$$

$$\xi = 14/9 \approx 1.556$$

Step 6: Verify $\xi \in (1, 2)$

$$1 < 14/9 < 2 \checkmark$$

Cauchy's MVT is verified!

4.4 Taylor's Theorem

Statement of the Theorem

If $f(x)$ has derivatives up to order n that are continuous on $[a, b]$ and the $(n+1)$ th derivative exists on (a, b) , then there exists at least one point $x = \xi \in (a, b)$ such that:

$$f(b) = f(a) + f'(a)/1! (b-a) + f''(a)/2! (b-a)^2 + \dots + f^{(n)}(a)/n! (b-a)^n + R_n$$

where R_n is the **remainder** (error term).

Forms of the Remainder

Lagrange's Form:

$$R_n = f^{(n+1)}(\xi)/(n+1)! \cdot (b-a)^{n+1}$$

This is the most commonly used form.

Cauchy's Form:

$$R_n = f^{(n+1)}(\xi)/n! \cdot (b-\xi)^n(b-a)$$

Integral Form:

$$R_n = 1/n! \int_a^b f^{(n+1)}(t)(b-t)^n dt$$

Taylor's Series (Expanded Form)

For a sub-interval $[x_0, x]$:

$$f(x) = f(x_0) + f'(x_0)/1! (x-x_0) + f''(x_0)/2! (x-x_0)^2 + \dots + f^{(n)}(x_0)/n! (x-x_0)^n + f^{(n+1)}(\xi)/(n+1)! (x-x_0)^{n+1}$$

Maclaurin Series

A Taylor series centered at $x_0 = 0$:

$$f(x) = f(0) + f'(0)/1! x + f''(0)/2! x^2 + f'''(0)/3! x^3 + \dots$$

Important Maclaurin Series

Function	Maclaurin Series	Interval
e^x	$1 + x + x^2/2! + x^3/3! + \dots$	All real x
$\sin x$	$x - x^3/3! + x^5/5! - x^7/7! + \dots$	All real x
$\cos x$	$1 - x^2/2! + x^4/4! - x^6/6! + \dots$	All real x
$\ln(1+x)$	$x - x^2/2 + x^3/3 - x^4/4 + \dots$	$-1 < x \leq 1$
$(1+x)^n$	$1 + nx + n(n-1)/2! x^2 + \dots$	$ x < 1$
$\tan^{-1}x$	$x - x^3/3 + x^5/5 - x^7/7 + \dots$	$ x \leq 1$

Example 1: Maclaurin Expansion

Problem: Find the Maclaurin series for $f(x) = e^x$.

Solution:

$$f(x) = e^x$$

$$f'(x) = e^x$$

$$f''(x) = e^x$$

$$f'''(x) = e^x$$

...

$$f^{(n)}(x) = e^x$$

At $x = 0$:

$$f(0) = 1$$

$$f'(0) = 1$$

$$f''(0) = 1$$

$$f'''(0) = 1$$

...

Maclaurin series:

$$e^x = 1 + x/1! + x^2/2! + x^3/3! + \dots$$

Example 2: Taylor Expansion

Problem: Expand $f(x) = \ln x$ about $x_0 = 1$.

Solution:

$$f(x) = \ln x$$

$$f'(x) = 1/x$$

$$f''(x) = -1/x^2$$

$$f'''(x) = 2/x^3$$

$$f^{(4)}(x) = -6/x^4$$

At $x_0 = 1$:

$$f(1) = 0$$

$$f'(1) = 1$$

$$f''(1) = -1$$

$$f'''(1) = 2$$

$$f^{(4)}(1) = -6$$

$$\ln x = (x-1) - (x-1)^2/2 + (x-1)^3/3 - (x-1)^4/4 + \dots$$

Example 3: Using Taylor's Theorem

Problem: Approximate $\ln(1.1)$ using Taylor's Theorem with $n = 3$.

Solution:

For $f(x) = \ln x$ about $x_0 = 1$:

$$\ln x \approx (x-1) - (x-1)^2/2 + (x-1)^3/3$$

For $x = 1.1$:

$$\begin{aligned}\ln(1.1) &\approx 0.1 - 0.01/2 + 0.001/3 \\ &= 0.1 - 0.005 + 0.000333 \\ &= 0.095333\end{aligned}$$

Actual value: $\ln(1.1) = 0.095310$ (accurate to 6 decimal places)

Error analysis: $R_3 = f^{(4)}(\xi)/4! \cdot (0.1)^4$

Since $f^{(4)}(x) = -6/x^4$, the maximum error occurs at the smallest x ($x = 1$):

$$|R_3| \leq 6/(1^4 \cdot 24) \cdot 0.0001 = 0.000025$$

So the approximation is accurate to about 5 decimal places.

4.5 Indefinite Integration

Definition

Indefinite integration is the reverse process of differentiation.

If:

$$d/dx [F(x)] = f(x)$$

Then:

$$\int f(x) dx = F(x) + C$$

where:

- **C** = constant of integration
- **f(x)** = integrand
- **F(x)** = antiderivative

Basic Integration Rules

Rule	Formula
Power Rule	$\int x^n dx = x^{n+1}/(n+1) + C, n \neq -1$
Natural Log	$\int 1/x dx = \ln x + C$
Constant Multiple	$\int k \cdot f(x) dx = k \cdot \int f(x) dx$
Sum/Difference	$\int [f(x) \pm g(x)] dx = \int f(x) dx \pm \int g(x) dx$

Special Cases

When $n = -1$ in Power Rule:

$$\int x^{-1} dx = \int 1/x dx = \ln|x| + C$$

Important: The formula $x^{n+1}/(n+1)$ fails when $n = -1$ because it gives division by zero!

Examples

Example 1:

$$\int x^4 dx = x^5/5 + C$$

Example 2:

$$\int 5x^6 dx = 5 \int x^6 dx = 5 \cdot x^7/7 + C = 5x^7/7 + C$$

Example 3:

$$\int x^{-3} dx = x^{-2}/(-2) + C = -1/(2x^2) + C$$

Example 4:

$$\int 1/x^5 dx = \int x^{-5} dx = x^{-4}/(-4) + C = -1/(4x^4) + C$$

Example 5:

$$\int 1/\sqrt{x} dx = \int x^{-1/2} dx = x^{1/2}/(1/2) + C = 2\sqrt{x} + C$$

Integration of Polynomials

Rule: Integrate each term separately.

Example:

$$\begin{aligned} \int (x^2 + 3x + 4) dx &= \int x^2 dx + 3 \int x dx + 4 \int 1 dx \\ &= x^3/3 + 3x^2/2 + 4x + C \end{aligned}$$

Functions Requiring Simplification

Example 1:

$$\begin{aligned} \int (x+1)^2 dx &= \int (x^2 + 2x + 1) dx \\ &= x^3/3 + x^2 + x + C \end{aligned}$$

Example 2:

$$\begin{aligned} \int (x^4 + 3x^3 + 4x^2)/x^4 dx \\ &= \int (1 + 3/x + 4/x^2) dx \\ &= x + 3\ln|x| - 4/x + C \end{aligned}$$

4.6 Integration that Gives Logarithms

The Rule

If the numerator is the **differential coefficient** of the denominator (apart from a constant), the integral is the **logarithm of the denominator**.

$$\int f'(x)/f(x) \, dx = \ln|f(x)| + C$$

Examples

Example 1:

$$\int (2x+1)/(2x^2+2x+7) \, dx$$

$$\text{Denominator: } u = 2x^2+2x+7$$

$$du/dx = 4x+2 = 2(2x+1)$$

$$(2x+1)dx = du/2$$

$$= 1/2 \int du/u = 1/2 \ln|u| + C$$

$$= 1/2 \ln|2x^2+2x+7| + C$$

Example 2:

$$\int 1/(3x+2) \, dx$$

$$= 1/3 \int 3/(3x+2) \, dx$$

$$= 1/3 \ln|3x+2| + C$$

Example 3:

$$\int x/(x^2+1) \, dx$$

$$\text{Denominator: } u = x^2+1$$

$$du/dx = 2x$$

$$x \, dx = du/2$$

$$= 1/2 \int du/u = 1/2 \ln|x^2+1| + C$$

4.7 Integration of Trigonometric Functions

Basic Integrals

Function	Integral
$\int \sin x \, dx$	$-\cos x + C$
$\int \cos x \, dx$	$\sin x + C$
$\int \tan x \, dx$	$-\ln \cos x + C$
$\int \cot x \, dx$	$\ln \sin x + C$
$\int \sec^2 x \, dx$	$\tan x + C$
$\int \csc^2 x \, dx$	$-\cot x + C$
$\int \sec x \cdot \tan x \, dx$	$\sec x + C$
$\int \csc x \cdot \cot x \, dx$	$-\csc x + C$

Derivation of $\int \tan x \, dx$

$$\int \tan x \, dx = \int \sin x / \cos x \, dx$$

$$\begin{aligned} \text{Let } u &= \cos x, \, du = -\sin x \, dx \\ \sin x \, dx &= -du \end{aligned}$$

$$\begin{aligned} &= \int -du/u = -\ln|u| + C \\ &= -\ln|\cos x| + C \end{aligned}$$

Integration of Squared Trig Functions

$\int \tan^2 x \, dx$:

$$\begin{aligned} \tan^2 x &= \sec^2 x - 1 \\ \int \tan^2 x \, dx &= \int \sec^2 x \, dx - \int 1 \, dx \\ &= \tan x - x + C \end{aligned}$$

$\int \cot^2 x \, dx$:

$$\begin{aligned} \cot^2 x &= \csc^2 x - 1 \\ \int \cot^2 x \, dx &= \int \csc^2 x \, dx - \int 1 \, dx \\ &= -\cot x - x + C \end{aligned}$$

$\int \sin^2 x \, dx$:

$$\begin{aligned} \sin^2 x &= (1 - \cos 2x)/2 \\ \int \sin^2 x \, dx &= 1/2 \int 1 \, dx - 1/2 \int \cos 2x \, dx \\ &= x/2 - \sin 2x/4 + C \end{aligned}$$

$\int \cos^2 x \, dx$:

$$\begin{aligned} \cos^2 x &= (1 + \cos 2x)/2 \\ \int \cos^2 x \, dx &= 1/2 \int 1 \, dx + 1/2 \int \cos 2x \, dx \\ &= x/2 + \sin 2x/4 + C \end{aligned}$$

Product-to-Sum Formulas

Use these when integrating products of sine and cosine:

$$\begin{aligned} \sin A \cos B &= 1/2[\sin(A+B) + \sin(A-B)] \\ \cos A \sin B &= 1/2[\sin(A+B) - \sin(A-B)] \\ \cos A \cos B &= 1/2[\cos(A+B) + \cos(A-B)] \\ \sin A \sin B &= 1/2[\cos(A-B) - \cos(A+B)] \end{aligned}$$

Example:

$$\int \sin 4x \cdot \sin 2x \, dx$$

Using formula:

$$\begin{aligned} \sin 4x \cdot \sin 2x &= 1/2[\cos(4x-2x) - \cos(4x+2x)] \\ &= 1/2[\cos 2x - \cos 6x] \end{aligned}$$

$$\begin{aligned}\int \sin 4x \cdot \sin 2x \, dx &= 1/2 \int \cos 2x \, dx - 1/2 \int \cos 6x \, dx \\ &= \sin 2x/4 - \sin 6x/12 + C\end{aligned}$$

4.8 Integration of Exponential Functions

Basic Rule

$$\int e^x \, dx = e^x + C$$

Compound Exponent

For $\int e^{(ax+b)} \, dx$:

Let $u = ax + b$, $du = a \, dx \rightarrow dx = du/a$

$$\int e^{(ax+b)} \, dx = 1/a \int e^u \, du = (1/a)e^u + C = (1/a)e^{(ax+b)} + C$$

Examples

Example 1:

$$\int e^{(3x)} \, dx = (1/3)e^{(3x)} + C$$

Example 2:

$$\int e^{(-2x-1)} \, dx = -(1/2)e^{(-2x-1)} + C$$

Example 3:

$$\int e^{(-x/2)} \, dx = -2e^{(-x/2)} + C$$

Example 4:

$$\int e^{(x^3)} \cdot x^2 \, dx$$

Let $u = x^3$, $du = 3x^2 \, dx$

$$x^2 \, dx = du/3$$

$$= 1/3 \int e^u \, du = (1/3)e^{(x^3)} + C$$

Definite Integrals with Exponentials

Example:

$$\int_1^2 e^{(2x+1)} \, dx$$

$$= [1/2 \cdot e^{(2x+1)}]_1^2$$

$$= 1/2[e^5 - e^3]$$

$$= 1/2 e^3(e^2 - 1)$$

4.9 Definite Integration

Definition

The definite integral of $f(x)$ from a to b is:

$$\int_a^b f(x) \, dx = [F(x)]_a^b = F(b) - F(a)$$

where $F(x)$ is any antiderivative of $f(x)$.

Properties of Definite Integrals

Property	Formula
Zero width	$\int_a^a f(x) \, dx = 0$
Reverse limits	$\int_a^b f(x) \, dx = -\int_b^a f(x) \, dx$
Constant multiple	$\int_a^b k \cdot f(x) \, dx = k \cdot \int_a^b f(x) \, dx$
Sum/difference	$\int_a^b [f(x) \pm g(x)] \, dx = \int_a^b f(x) \, dx \pm \int_a^b g(x) \, dx$
Additivity	$\int_a^b f(x) \, dx + \int_b^c f(x) \, dx = \int_a^c f(x) \, dx$

Examples

Example 1:

$$\begin{aligned} & \int_0^1 (x^3 - 3x^2 + 5) \, dx \\ &= [x^4/4 - x^3 + 5x]_0^1 \\ &= (1/4 - 1 + 5) - (0 - 0 + 0) \\ &= 1/4 + 4 \\ &= 17/4 \end{aligned}$$

Example 2:

$$\begin{aligned} & \int_4^6 (\sqrt{x} + 1/\sqrt{x} + 2) \, dx \\ &= [2x^{(3/2)}/3 + 2\sqrt{x} + 2x]_4^6 \\ &= [2 \cdot 6^{(3/2)}/3 + 2\sqrt{6} + 12] - [2 \cdot 4^{(3/2)}/3 + 2 \cdot 2 + 8] \\ &= [2 \cdot 6\sqrt{6}/3 + 2\sqrt{6} + 12] - [2 \cdot 8/3 + 4 + 8] \\ &= [4\sqrt{6} + 2\sqrt{6} + 12] - [16/3 + 12] \\ &= 6\sqrt{6} + 12 - 16/3 - 12 \\ &= 6\sqrt{6} - 16/3 \end{aligned}$$

Practice Problems and Solutions

Problem 1

Verify Rolle's Theorem for $f(x) = x^2 - 4x + 3$ on $[1, 3]$.

Solution:

$$\begin{aligned}f(1) &= 1 - 4 + 3 = 0 \\f(3) &= 9 - 12 + 3 = 0 \\f(1) &= f(3) \checkmark\end{aligned}$$

$$\begin{aligned}f'(x) &= 2x - 4 \\f'(\xi) &= 0 \rightarrow 2\xi - 4 = 0 \rightarrow \xi = 2\end{aligned}$$

$$\xi = 2 \in (1, 3) \checkmark$$

Problem 2

Verify the MVT for $f(x) = x^3 - 2x$ on $[0, 2]$.

Solution:

$$\begin{aligned}f(0) &= 0 \\f(2) &= 8 - 4 = 4\end{aligned}$$

$$[f(2) - f(0)]/(2 - 0) = 4/2 = 2$$

$$\begin{aligned}f'(x) &= 3x^2 - 2 \\f'(c) &= 2 \rightarrow 3c^2 - 2 = 2 \rightarrow 3c^2 = 4 \rightarrow c = 2/\sqrt{3}\end{aligned}$$

$$c = 2/\sqrt{3} \approx 1.155 \in (0, 2) \checkmark$$

Problem 3

Find the Maclaurin series for $\sin x$.

Solution:

$$\begin{aligned}f(x) &= \sin x \\f'(x) &= \cos x \\f''(x) &= -\sin x \\f'''(x) &= -\cos x \\f^{(4)}(x) &= \sin x\end{aligned}$$

At $x = 0$:

$$\begin{aligned}f(0) &= 0 \\f'(0) &= 1 \\f''(0) &= 0 \\f'''(0) &= -1 \\f^{(4)}(0) &= 0\end{aligned}$$

$$\sin x = x - x^3/3! + x^5/5! - x^7/7! + \dots$$

Problem 4

Evaluate $\int (x + 1/x)^2 dx$

Solution:

$$(x + 1/x)^2 = x^2 + 2 + 1/x^2$$

$$\begin{aligned}\int (x + 1/x)^2 dx &= \int x^2 dx + \int 2 dx + \int x^{-2} dx \\ &= x^3/3 + 2x - 1/x + C\end{aligned}$$

Problem 5**Evaluate $\int_0^1 \sqrt{x}(1 - x^3)^2 dx$** **Solution:**

$$\begin{aligned}\sqrt{x}(1 - x^3)^2 &= x^{1/2}(1 - 2x^3 + x^6) \\ &= x^{1/2} - 2x^{7/2} + x^{13/2}\end{aligned}$$

$$\begin{aligned}&\int_0^1 x^{1/2} dx - 2\int_0^1 x^{7/2} dx + \int_0^1 x^{13/2} dx \\ &= [2x^{3/2}/3]_0^1 - 2[2x^{9/2}/9]_0^1 + [2x^{15/2}/15]_0^1 \\ &= 2/3 - 4/9 + 2/15 \\ &= 30/45 - 20/45 + 6/45 \\ &= 16/45\end{aligned}$$

Problem 6**Evaluate $\int(\sqrt{x^5} - \sqrt[5]{x}) dx$** **Solution:**

$$\begin{aligned}\sqrt{x^5} &= x^{5/2} \\ \sqrt[5]{x} &= x^{1/5}\end{aligned}$$

$$\begin{aligned}&\int x^{5/2} dx - \int x^{1/5} dx \\ &= x^{7/2}/(7/2) - x^{6/5}/(6/5) + C \\ &= 2x^{7/2}/7 - 5x^{6/5}/6 + C\end{aligned}$$

Problem 7**Evaluate $\int e^{(-5\theta+1/3)} d\theta$** **Solution:**

$$\begin{aligned}\text{Let } u &= -5\theta + 1/3 \\ du/d\theta &= -5 \\ d\theta &= -du/5\end{aligned}$$

$$\int e^u (-du/5) = -1/5 e^u + C = -1/5 e^{(-5\theta+1/3)} + C$$

Problem 8

Evaluate $\int_0^{\pi/2} \sin^2 x \, dx$

Solution:

$$\sin^2 x = (1 - \cos 2x)/2$$

$$\begin{aligned}\int_0^{\pi/2} \sin^2 x \, dx &= 1/2 \int_0^{\pi/2} 1 \, dx - 1/2 \int_0^{\pi/2} \cos 2x \, dx \\ &= 1/2 [x]_0^{\pi/2} - 1/2 [\sin 2x/2]_0^{\pi/2} \\ &= 1/2 (\pi/2) - 1/4 [\sin \pi - \sin 0] \\ &= \pi/4 - 0 \\ &= \pi/4\end{aligned}$$

Common Mistakes and How to Avoid Them

Mistake 1: Forgetting the Constant of Integration

$$\begin{aligned}\int x^2 \, dx &= x^3/3 && \times \\ \int x^2 \, dx &= x^3/3 + C && \checkmark\end{aligned}$$

Mistake 2: Using Power Rule for $n = -1$

$$\begin{aligned}\int 1/x \, dx &= x^0/0 + C && \times \\ \int 1/x \, dx &= \ln|x| + C && \checkmark\end{aligned}$$

Mistake 3: Not Checking Conditions for MVT

- Must verify continuity and differentiability first!

Mistake 4: Confusing Taylor and Maclaurin Series

- Taylor about ANY point
- Maclaurin specifically about $x = 0$

Mistake 5: Forgetting Absolute Values in Log Integrals

$$\begin{aligned}\int 1/x \, dx &= \ln x + C && \times \\ \int 1/x \, dx &= \ln|x| + C && \checkmark\end{aligned}$$

Key Formulas Summary

Rolle's Theorem

f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$
 $\rightarrow \exists \xi \in (a,b): f'(\xi) = 0$

First MVT

f continuous on $[a,b]$, differentiable on (a,b)
 $\rightarrow \exists \xi \in (a,b): f'(\xi) = [f(b)-f(a)]/(b-a)$

Second MVT (Cauchy)

f, g continuous on $[a, b]$, differentiable on (a, b)

$\rightarrow \exists \xi \in (a, b): f'(\xi)/g'(\xi) = [f(b)-f(a)]/[g(b)-g(a)]$

Taylor's Theorem

$$f(x) = \sum f^{(n)}(a)/n! (x-a)^n + R_n$$

Integration Rules

$$\int x^n dx = x^{n+1}/(n+1) + C \quad (n \neq -1)$$

$$\int 1/x dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$